



Data Mining

Assignment 1

Getting to Know Your Data

Instructor:

Dr. Mostafa Fakhrahmad

Deadline:

Friday, 18 Aban 1403, at 11:59 PM (Tehran Time)

Introduction

In this assignment, you will perform data exploration on a relational dataset. Data exploration involves an initial examination of the dataset to gain insights into its characteristics. Engaging in data exploration can help you determine the most suitable preprocessing and analysis methods to be used in the later stages of data mining.

The assignment begins with an introduction to the Iris dataset, followed by tasks that involve calculating univariate and bivariate statistics to gain insights into the data. Next, various visualizations will be created to further explore the relationships between the attributes. Finally, the assignment concludes with a review of the deliverables, submission guidelines, and academic integrity policies to ensure a thorough understanding of the expectations.

The Iris Dataset

The Iris dataset is a classic dataset in the field of machine learning and statistics, introduced by Ronald Fisher in 1936 [1]. It contains measurements of 150 iris



flowers, each belonging to one of the following species: *Iris setosa*, *Iris versicolor*, and *Iris virginica*. Each flower is characterized by five attributes:

- Sepal length (in centimeters)
- Sepal width (in centimeters)
- Petal length (in centimeters)
- Petal width (in centimeters)
- Class label, which specifies the species (*I. setosa*, *I. versicolor*, *I. virginica*)

The first four attributes are ratio-scaled, while the last one is a nominal attribute according to Stevens' typology [2].

You will need to download the Iris dataset for this assignment. One way to obtain it is through the University of California at Irvine (UCI) Machine Learning Repository. Please note that when referring to a class in your answers, use only the species names: **setosa**, **versicolor**, and **virginica**.

1 Univariate Summary Statistics

For the **sepal width** attribute of the dataset, you are required to calculate the following statistics *for each species separately*:

1. Number of missing values
2. Minimum
3. First quartile (Q_1)
4. Median
5. Third quartile (Q_3)
6. 95th percentile
7. Maximum
8. Mean
9. Range
10. Interquartile range (IQR)



11. Sample standard deviation
12. Population standard deviation
13. Median absolute deviation (MAD)

We would like to reiterate that these statistics should be calculated *individually for each class*. When calculating statistics, please ensure that any missing values are excluded from your computations, if they exist. Please save the results in a file named `statistics.csv`. The first line (header) of the file should be as follows:

```
label,missing,min,q1,med,q3,p95,max,mean,range,iqr,std,std_pop,mad
```

2 Correlation Analysis

In this section, you will perform a simple correlation analysis on the Iris dataset. Specifically, you will create a correlation matrix for all attributes except the label as follows:

1. Using Pearson's correlation coefficient, compute the correlation for every pair of attributes, *excluding the label*, in a correlation matrix.
2. From the correlation matrix, identify the pair of distinct features that exhibit the minimum absolute correlation. Do the same for the maximum absolute correlation.
3. Interpret the results from the previous step.

Please save your correlation matrix in a CSV file named `correlations.csv`. The file should not have any headers or indices (primary keys) and should only include four columns and four rows. The rows and columns should implicitly represent the following attributes in the specified order: sepal length, sepal width, petal length, and petal width.

3 Visualization

In this section, you are asked to create several plots based on the Iris dataset. The attributes of focus in this section are **petal length** and **sepal width**, unless otherwise specified.

1. **Distribution of Labels:** Create a bar plot to show the distribution of species labels. Is the label distribution balanced? Explain.



2. Histograms:

- Create separate histograms for each of the two features.
- Create a single 3D histogram for these two features. The axes of the plot should represent petal length, sepal length, and count.

3. Box Plots:

- Create a box plot for each of the two features (petal length and sepal width). Each box plot should contain three distinct boxes, with each box representing the distribution of the selected feature for data items *belonging to a specific species class*.
- Create a two-dimensional box plot showing the distribution of one feature against the other. There is no need to draw separate boxes for each class.
- In your explanation, describe the meaning of the whiskers, how potential outliers are identified, represented, or ignored, and any other, and any other relevant details about the box plots' configuration.

4. Quantile Plots:

Generate a quantile plot for each of the two features. Provide an explanation of how you generated the plots and what the plots indicate.

5. Scatter Plots:

- Use **all four quantitative features** for this part: Create six 2D scatter plots to visualize the relationship between every distinct feature pair. In the plots, *differentiate the points by color or shape for each species*.
- Based on the results from the previous step, consider the task of species classification. Suppose you are using exactly two features to predict the species of a given iris. Compare the following two scenarios for feature selection: using sepal length and sepal width versus using sepal length and petal length. Which feature pair would allow for better distinction of the species class? Justify your answer.
- Create a 3D scatter plot using **sepal length**, **sepal width**, and **petal length**. *Differentiate the points by color or shape for each species class*. Ensure that the plot is presented from a good viewing angle to better visualize separability among the species. Based on this visualization, assess whether these three features are fairly sufficient for classifying the data items into their respective species classes.

6. Probability Distributions:



- Estimate and plot the probability density function (PDF) for **petal length** *grouped by species*, i.e. in the same plot, draw three curves, each representing the distribution of petal length for data items in the same species class. Explain the method used for estimating the PDF.
- Use this plot to identify classification rules that can be generally applied to determine the species of an iris based on its petal length.

Deliverables

You are required to deliver three sets of items:

1. Code Workspace:

- This should contain all your code related to the assignment.
- Use the Python programming language for your code.
- Ensure that your code is well organized and easy to understand.

2. CSV Files: Include `statistics.csv` and `correlations.csv` relevant to your analysis in Section 1 and Section 2, respectively.

3. PDF Report:

- It may be written in Persian or English.
- Include a detailed explanation of what you did for each step of the assignment in the report.
- Ensure the report is well-structured and in a clear and readable format.
- Include the results obtained from each step, including plots.
- Round the numerical results in your report, if present, to three decimal places.
- You do *not* need to re-represent the results of the CSV files in your report. If you do, ensure a readable format.
- Throughout your analysis, pay attention to any interesting observations or patterns you notice in the data. You are encouraged to discuss these insights in your report.

Assuming your name is Jane Doe and your student number is 90012345, you need to create a directory named `DM-HW1-Jane_Doe-90012345` as the root of your workspace. You then need to structure your deliverables as follows:



```
DM-HW1-Jane_Doe-90012345/  
|--- src/  
|       |--- run.py  
|--- dist/  
|       |--- correlations.csv  
|       `--- statistics.csv  
|--- data/  
|--- requirements.txt  
`--- DM-HW1-Jane_Doe-90012345-Report.pdf
```

In this structure:

- Your report should be in the file `DM-HW1-Jane_Doe-90012345-Report.pdf`.
- Put the names of utilized libraries, if any, in the `requirements.txt` file, one library per line.
- Inside the `src` directory, place your Python source files. The main source file should be named `run.py`. To keep your code organized, you may have additional source files as well, but the one that we will run to see your results is `run.py`.
- If you read the dataset from a file, place the dataset file or files into the `data` directory.
- When `run.py` is executed, it should:
 - Read the dataset from the `data` directory *or* download it from the web.
 - Create or overwrite the files `correlations.csv` and `statistics.csv` in the `dist` directory.
 - Display the plots *or* save them to the `dist` directory.
- When writing paths to input or output files, always use relative paths instead of absolute paths. This ensures that your code can be executed on different machines without modification, making it easier for us to run your code successfully!

Submission Guidelines

Create a ZIP file named `DM-HW1-Jane_Doe-90012345.zip` from your workspace. Then, email it to `data.mining.graders@gmail.com`. The subject of the email should



be Assignment 1: 90012345, replacing 90012345 with your student number. The deadline is **Friday, 18 Aban 1403, at 11:59 PM (Tehran Time)**.

Make sure to double-check that all required files are included in the workspace and that your code runs without errors before submission. Late submissions may incur a penalty.

Academic Integrity Guidelines

1. **Independent Work:** While you are encouraged to discuss the questions with your peers, please write the code and answers independently. Do not copy others' answers or code.
2. **Use of Resources:** If you use online code resources or generative AI tools (such as ChatGPT, Claude, Copilot, Llama, etc.) for any part of your assignment, please document that usage in your report. Be specific about what resources you used and how they contributed to your work.
3. **Originality:** Your submission should reflect *your* understanding of the material. Plagiarism or submitting work that is not your own will not be tolerated.
4. **Ask for Help:** If you are struggling with a concept, feel free to ask for help from your instructor or graders. Seeking clarification is encouraged, but ensure that your final submission is your own work.

References

- [1] R. A. Fisher, "The use of multiple measurements in taxonomic problems," *Annals of Eugenics*, vol. 7, no. 2, pp. 179–188, Sep. 1936, ISSN: 2050-1439. DOI: 10.1111/j.1469-1809.1936.tb02137.x.
- [2] S. S. Stevens, "On the theory of scales of measurement," *Science*, vol. 103, no. 2684, pp. 677–680, Jun. 1946, ISSN: 1095-9203. DOI: 10.1126/science.103.2684.677.